INCH-POUND

MIL-DTL-24640/3D
22 August 2002
SUPERSEDING
MIL-C-24640/3C
26 June 1995

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT-WEIGHT, FOUR-CONDUCTOR, LOW POWER, TYPE FX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24640 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

Construction (Non-Watertight)

First - 19-strand, tin-coated copper conductor, and AWG size and dimensions in accordance with Table I.

Second - Component wires shall be MIL-W-81044/12, except identification shall be in accordance with the codes and methods of MIL-DTL-24640. Testing shall be done in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with MIL-W-81044, prior to cabling. Identification codes applied using Method 1 or 3. If method 1 is used, component wire shall be white. Dimensions as shown in Table I.

Third - Four wires shall be cabled together.

Fourth - Binder tape applied helically with overlap, at manufacturer's option.

Fifth - Crosslinked polyolefin jacket. (See Table I for thickness).

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Construction details.

PIN M24640/3	Type Number	AWG Size	Conductor diameter (inch) (nominal)	Component wire diameter (inch) (min)	(max)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)	Conductor resistance per 1000 feet (ohms)
-01UN	FX-3	16	0.056	0.069	0.075	0.031	0.262	0.282	63.0	4.98
-02UN	FX-4	14	.071	.085	.093	.034	.311	.335	92.6	3.18

REQUIREMENTS:

Qualification required.
Construction details: See Table I

Component wire: The component wire shall be per MIL-W-81044 and shall be tested in accordance with the following wire ratings and additional requirements.

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 300 ± 2°C (572 ± 3.6°F) for 6 hours.
Blocking: 150 ± 2°C (302 ± 3.6°F)
Color: In accordance with MIL-STD-104, class 1; white preferred
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test:
Primary insulation (when tested in lieu of spark test):
6.0 kilovolts (peak) 100 percent test.
Component wire: 8.0 kilovolts (peak), or 3 kHz spark test, 5.5 kV, 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200 ± 2°C (392 ± 3.6°F) for 168 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300 ± 2°C (572 ± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz or 3 kHz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	See Table I
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	Not applicable
Shield to shield	Not applicable
Insulation resistance – megohms-1000 feet, minimum	5000

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Conductor continuity ..... No failure
Jacket flaws ..... Not applicable

Group A:
Visual and dimensional ..... No failure
Capacitance:
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum ..... Not applicable
Capacitance unbalance - percent, maximum ..... Not applicable
Characteristic impedance - at 1 MHz, ohms ..... Not applicable
Watertightness (see Table IX of MIL-DTL-24640 for limits of water leakage) ..... Not applicable
Weight ..... No failure

Group B:
Cold bending ..... No failure
Cross link proof test (percent maximum)
Jacket 50
Drip ..... Not applicable
Tear - pounds per inch thickness, minimum 35
Physicals (unaged)
Jacket
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160
Attenuation - at 3 MHz, dB/100 feet, maximum ..... Not applicable
Gas flame (1 hour) ..... Not applicable

Group C:
Physicals (aged)
Jacket
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged, minimum 60
Permanence of printing (conductor) - cycles, method 1only, minimum 25
Permanence of printing (jacket) - cycles, minimum 125
Shield - conformance to material, construction and coverage ..... Not applicable
Heat distortion - percent of unaged, maximum 30
Cable sealant removability ..... Not applicable
Shrinkage ..... No failure
Mutual inductance - dB/1000 feet, maximum ..... Not applicable
Surface transfer impedance
milliohms per meter, maximum ..... Not applicable
EMP response, dB, minimum ..... Not applicable
Pulse response time - ms/500 feet
Rise time, maximum ..... Not applicable
Decay time, maximum ..... Not applicable

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Flame propagation	No failure
Aging and compatibility cable	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure
Fungus Resistance	No failure
Abrasion resistance	No failure
Gas Flame (3 hour)	Not applicable
Component wire	No failure

Unit ordering length: 500 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2307-003)

Review Activities:
Army - AR, AV, CR, CR4
Navy - CG, EC
DLA - IS

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